ML TEAM — SPRING 2026

Practice Worksheet: Neural Networks · CNNs · Transformers · Attention · RL · Q-Learning

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Section 1 — Neural Networks (/ 20 pts)

Q1 **[Calculation]** A neuron has weights w = [3, -2, 1], bias b = -1, and input x = [2, 1, 4]. Compute the raw output z = w·x + b, then apply ReLU. Show your work.

$z = 8 - 1 = 7$

$ReLU(z) = \max(0, z) = 7$

Q2 **[Short Answer]** A network has 3 layers with dimensions: input (784) → hidden (128) → hidden (64) → output (10). How many learnable parameters are in the weight matrices only (ignore biases)?

$784 \times 128 + 128 \times 64 + 64 \times 10 = 109184$ learnable params

Q3 **[Explain]** Without activation functions, a 10-layer neural network collapses to a single linear transformation. Explain why using matrix algebra.

Matrix multiplication is a linear transformation Concatenating a linear transformation 10 x remains linear. A non-linear element must be introduced to create non-linearity

Q4 **[Short Answer]** You're training a classifier and notice the training loss is very low but validation loss is very high. Name this problem, and give two ways to fix it using what you know about neural networks.

Overfitting - Randomly dropout neurons, L1/L2 regularization

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Section 2 — Convolutional Neural Networks (/ 20 pts)

Q5 [Calculation] Apply the 1D filter W = [1, 0, -1] with stride 1 to input X = [3, 1, 4, 1, 5, 9]. Write the full output and explain what pattern this filter detects.

X=[3 1 4 1 5 9] [-1, 0, -1, -8] = Output

W=[1 0 -1]

W=[1 0 -1]

W=[1 0 -1]

W=[1 0 -1]

Pattern: Edge detector, it measures rate of change ~~of~~ across sequence.

Q6 [Calculation] An image is 64×64. You apply: (1) Conv with 3×3 filter, stride 1, no padding. (2) 2×2 max pool, stride 2. (3) Conv with 3×3 filter, stride 1, no padding. What is the spatial size after each step?

(1) $O = \frac{64-3+0}{1} + 1 = 62$ $O = \left\lfloor \frac{W-K+2P}{S} \right\rfloor + 1$

(2) $O = \frac{62-2+0}{2} + 1 = 31$

(3) $O = \frac{31-3+0}{1} + 1 = 29$

Q7 [Explain] Why do early CNN layers detect edges while deep layers detect semantic concepts like 'face' or 'wheel'? Explain in terms of receptive fields.

Neurons in early layers have a very small receptive field so they can ~~at~~ only detect simple features like edges, corners. Pooling

Q8 [Compare] A fully connected layer connecting 64×64 input to 1000 neurons has how many weights? A convolutional layer with 32 filters of size 3×3 on the same input has how many weights? Why is this difference important?

a. 64×64 ×1000 = 4096 000 weights

b. 3×3 ×32 =288 weights,

It is important b.c. CNN allows us to learn the information in graphics with much less weights, which means faster training and smaller storage.

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Section 3 — Transformers & Attention (/ 20 pts)

Q9 [Explain] Consider: 'I saw the man with the telescope.' Explain how attention assigns different weights to 'man' and 'telescope' when computing the contextual embedding for 'with'. What does this tell us about the sentence's ambiguity?

Product between query vector & key vector. The key vector for man & telescope is dot product with query vector of with. This is different because of their different positions in the sentence & different gramatical use in the sentence and have different embeddings themselves as words.

Q10 [Calculation] You have Q = [1, 0], K = [[1, 0], [0, 1], [-1, 0]], d_k = 2. Compute the raw attention scores, scale by 1/√d_k, and apply softmax. Which key does Q attend to most?

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) = \text{softmax}\left(\frac{[1\ \ 0]_{1\times 2}\begin{bmatrix}1 & 0 & -1\\ 0 & 1 & 0\end{bmatrix}_{2\times 3}}{\sqrt{2}}\right)$$

$$\begin{matrix}1\\0\end{matrix}_{2\times 1} \quad \begin{matrix}1 & 0 & -1\\ 0 & 1 & 0\end{matrix}_{2\times 3}$$

$$10_{1\times 2}\ 2\times 3$$

$$= \text{softmax}\left(\frac{[1\ \ 0\ \ -1]}{\sqrt{2}}\right) = \frac{[e^{1/\sqrt{2}}\ \ e^{0}\ \ e^{-1/\sqrt{2}}]}{e^{1/\sqrt{2}} + e^{0} + e^{-1/\sqrt{2}}}$$

First key since largest val

Q11 [Short Answer] A transformer is trained on the sentence 'I love to eat french fries with ketchup.' During training it predicts every word autoregressively. Draw the causal mask as a 9×9 grid (use 1 = attend, 0 = masked) for the first 4 tokens.

$$\begin{bmatrix}1 & 1 & 1 & 1\\ 0 & 1 & 1 & 1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix}$$

I love to eat

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\ & & & & & & 0 & 0 & 0\\ & & & & & & & & 0\\ & & & & & & & & 0\end{bmatrix}$$

Q12 [Compare] BERT uses bidirectional attention. GPT uses causal (left-to-right) attention. For each model, explain which task it is better suited for and why the masking choice directly causes this.

BERT is better to understand context & bi-directional meaning
GPT is better at text generation, coherent text, unidirectional.
Masking prevents the model from cheating by looking ahead (GPT)
both-direct (BERT)

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Section 4 — Reinforcement Learning (/ 20 pts)

Q13 **[Calculation]** $\gamma = 0.8$. An agent receives rewards $r_0=0$, $r_1=0$, $r_2=5$, $r_3=0$, $r_4=10$ then the episode ends. Compute the discounted return G_0 starting from t=0.

$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 r_{t+3} + \gamma^4 \cdot r_4$

$G_0 = 0 + 0.8 \cdot 0 + 0.8^2 \cdot 5 + 0.8^3 \cdot 0 + 0.8^4 \cdot 10$
$= 7.296$

Q14 **[Short Answer]** Define the Markov property. Give one example of a real-world problem where the Markov property holds, and one where it does NOT hold.

Markov property means that the future is only dependent on the current state.
Hold: In a chess game the future state depends only on the current state.
NOT hold: in a stock market, markov property does not usually hold.

Q15 **[Explain]** What is the credit assignment problem? Use a chess analogy. How does the Bellman equation help solve it?

RL algorithm needs to assign credit to every historical step, not just the current step. If step 10 leads to a win in a chess game, we need to assign credit to step 0, 1, 2, ..., 10.
Bellman Equation solves it by breaking the value of a state down into immediate reward $V(s) = R(s,a) + \gamma V(s')$

Q16 **[Compare]** An agent uses ε-greedy with ε = 0.1. (a) What fraction of actions are random? (b) If the agent is near a cliff with -100 reward, what is the expected penalty per step from random actions? (c) Should ε be higher early or late in training? Why?

a) 10% of actions are random.
b) $E = 0.1 \times -100 = -10$ per step.
c) ε should be higher in early training. A high ε forces the agent to try many different actions to discover which ones leads to high rewards.

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Ep2 Step3 (S2→S3)	$S_2 \to S_3$	20	$10 + 0.5\,(20 + 1.0 - 10)$	5	15

Q19 [Short Answer] Q-learning is called 'off-policy.' What does this mean? Write out the SARSA update rule and explain the one key difference from Q-learning's update rule.

It means that Q-learning learns the optimal policy independently of the agents actual behavior.

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)]$

SARSA is an on-policy update rule.

Q20 [Explain] A Q-table has 10,000 states and 5 actions. Describe one situation where this is fine and one where this is completely unworkable. What is the standard fix for the unworkable case?

The setup is fine for a simple 2D grid state world.

It is unworkable for high dimensions, or scenorios where states continuous (e.g. robotics, video games)

Deep Q-learning can help us fix the unworkable case.

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